

Week 3 Internship Tasks

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Organization : InterNova

Domain : Data Analytics

Task 1

Source Code:

```
create database student_data;
use student_data;
create table Students(
std_name varchar(50),
rollno int,
age int,
marks decimal(3,1),
grade char(1));

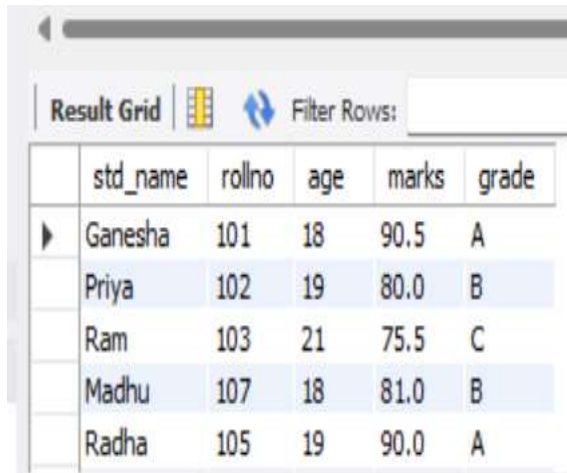
insert into Students(std_name,rollno,age,marks,grade)
values("Ganesha",101,18,90.5,'A'),
("Priya",102,19,80,'B'),
("Ram",103,21,75.5,'C'),
("Madhu",107,18,81,'B'),
("Radha",105,19,90,'A'),
("Meera",108,18,92,'A');
select * from Students;
select std_name,rollno,marks from Students;
```

Task 1

Source Code:

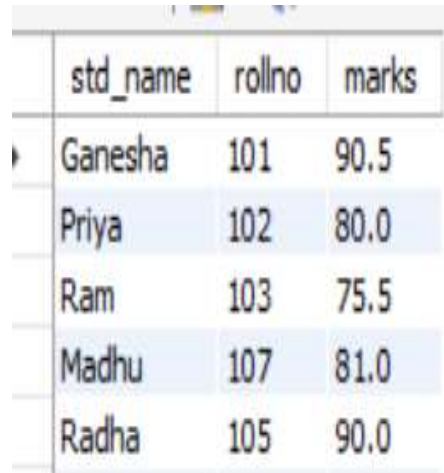
```
select std_name as student_name,marks as Scorefrom Students;
```

Output:



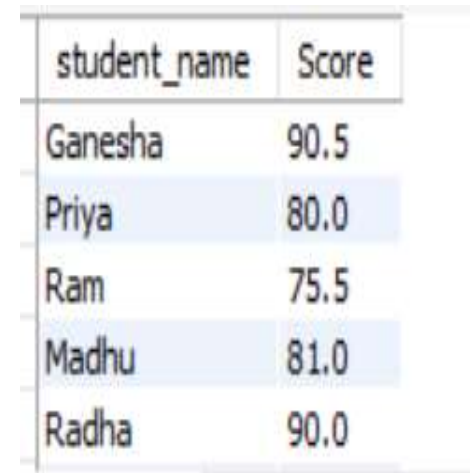
A screenshot of a database application's 'Result Grid'. It features a toolbar with a 'Filter Rows' button and a search input field. The table below displays student information with columns for name, roll number, age, marks, and grade.

	std_name	rollno	age	marks	grade
▶	Ganesha	101	18	90.5	A
	Priya	102	19	80.0	B
	Ram	103	21	75.5	C
	Madhu	107	18	81.0	B
	Radha	105	19	90.0	A



A partial screenshot of the result grid, showing only the first three columns: std_name, rollno, and marks.

	std_name	rollno	marks
▶	Ganesha	101	90.5
	Priya	102	80.0
	Ram	103	75.5
	Madhu	107	81.0
	Radha	105	90.0



Another partial screenshot of the result grid, showing only the last two columns: student_name and Score.

	student_name	Score
	Ganesha	90.5
	Priya	80.0
	Ram	75.5
	Madhu	81.0
	Radha	90.0

Task 1

Brief Explanation:

- 1) Created and used a sample database table.
- 2) Retrieved complete table data using `SELECT *`.
- 3) Selected required columns using `SELECT`.
- 4) Used column aliases with `AS` for clear and meaningful column names.
- 5) Practiced basic SQL data retrieval.

Task 2

Source Code:

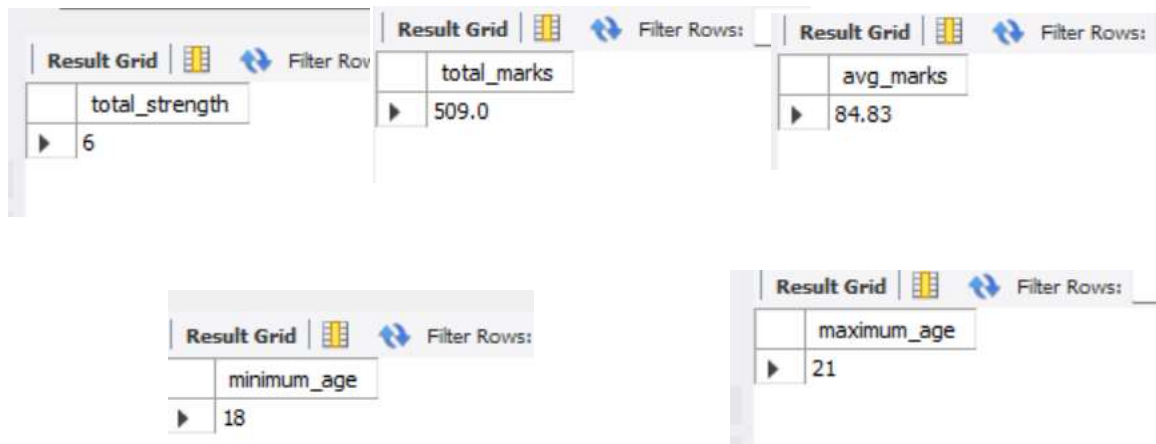
```
use student_data;
select * from students;
#Filter records using the WHERE clause.
select std_name,age from students where age=18;
select std_name,grade from students where grade between 'A' and 'C';
# Use comparison operators in conditions.
select std_name,marks from students where marks<=80;
# Sort records using ORDER BY.
select * from students order by marks desc;
select std_name,age from students order by age;
#Perform calculations using aggregate functions:
select count(*) as total_strength from students;
select sum(marks) as total_marks from students;
select round(avg(marks),2) as avg_marks from students;
select min(age) as minimum_age from students;
select max(age) as maximum_age from students;
```

Task 2

Source Code:

```
select std_name as student_name,marks as Scorefrom Students;
```

Output:



The image displays four separate screenshots of a SQL query result grid, each showing a different aggregate function result. Each grid has a header row with the column name and a single data row below it. The grids are arranged in two rows of two.

total_strength
6

total_marks
509.0

avg_marks
84.83

minimum_age
18

maximum_age
21

Brief Explanation:

- Filtered and sorted records.
- Used comparison operators.
- Applied COUNT(), SUM(), AVG(), MIN(), and MAX().
- Displayed query results

Task 3

Source Code:

```
select * from students;
#Group records using GROUP BY.
#Counting number of students are in each grade:

select grade,count(std_name) as count from studentsgroup by grade;

# Aggregate value for each group#
Find the average marks for each gradeselect grade,
round(avg(marks),2) as average_marks
from studentsgroup by grade;

#Use the HAVING clause to filter grouped results.
# display only grades where the average marks are above 85

select grade,avg(marks) as avg_marks from students
group by grade having avg_marks>85;
```

Task 3

Output:

	grade	average_marks
▶	A	90.83
	B	80.50
	C	75.50

	grade	avg_marks
▶	A	90.83333

	grade	count
▶	A	3
	B	2
	C	1

Brief Explanation:

- Grouped records using GROUP BY.
- Calculated aggregate values for each group.
- Filtered grouped results using HAVING.
- Displayed the results.

Task 4

Source Code:

```
select * from students;  
use student_data;
```

#Courses table

```
create table courses(  
rollno int,  
course varchar(50),  
department varchar(50));  
insert into courses(rollno,course,department)  
values(101,"AI","CSE"),  
(102,"DS","ECE"),  
(103,"Python","EEE"),  
(104,"Java","Mech"),  
(105,"ML","CSE"),  
(106,"AIML","ECE");  
select * from courses;
```

Task 4

Source Code:

#Inner Join

```
select s.std_name ,s.rollno,c.course,c.department from students  
as s inner join courses as c  
on s.rollno=c.rollno;
```

#left join

```
select s.std_name,s.rollno,c.course from students as s  
left join courses as c  
on s.rollno=c.rollno;
```

#Right join

```
select s.std_name,c.course from students as s  
right join courses as c  
on s.rollno=c.rollno;
```

Task 4

Output:

	std_name	rollno	course	department
▶	Ganesha	101	AI	CSE
	Priya	102	DS	ECE
	Ram	103	Python	EEE
	Radha	105	ML	CSE

	std_name	rollno	course
▶	Ganesha	101	AI
	Priya	102	DS
	Ram	103	Python
	Madhu	107	NULL
	Radha	105	ML

	std_name	course
▶	Ganesha	AI
	Priya	DS
	Ram	Python
	NULL	Java
	Radha	ML

Brief Explanation:

- Used two related tables with a common column.
- Applied INNER JOIN, LEFT JOIN, and RIGHT JOIN.
- Compared the results of each join.

Task 5

Source Code:

#Subqueries

#Find students scoring above the average marks

```
select * from students;  
select std_name,marks from students  
where marks >(select avg(marks) as avg_marks from students);
```

#Find students who have the highest marks

```
select std_name,marks from students  
where marks = (select max(marks) from students);
```

Task 5

Output:

	std_name	marks
▶	Ganesha	90.5
	Radha	90.0
	Meera	92.0

	std_name	marks
▶	Meera	92.0

Brief Explanation:

- 1)Used subqueries to solve two query problems
- 2)Retrieved result based on another query's result

Task 6

Output:

task6 - Excel (Product Activation Failed)													
File Home WPS PDF Insert Page Layout Formulas Data Review View Tell me what you want to do...													
Clipboard Font Alignment Number Styles Cells Editing Library AI Tools Output Conversion WPS PDF													
N7													
	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order_Date	Customer_ID	Customer_Name	Product_ID	Product	Category	Region	Quantity	Unit_Price	Sales	Payment_Method	Order_Status	
2	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	Card	Cancelled	
3	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	Net Banking	Cancelled	
4	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	Card	Cancelled	
5	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	Net Banking	Cancelled	
6	01-06-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	Card	Cancelled	
7	01-07-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	Net Banking	Cancelled	
8	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	Card	Cancelled	
9	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	Net Banking	Cancelled	
10	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	Card	Cancelled	
11	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	Net Banking	Cancelled	
12	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	Card	Delivered	
13	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	Net Banking	Delivered	
14	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	Card	Delivered	
15	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	Net Banking	Delivered	
16	08-05-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	Card	Delivered	
17	07-06-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	Net Banking	Delivered	
18	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	Card	Delivered	
19	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	Net Banking	Delivered	
20	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	Card	Delivered	
21	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	Net Banking	Delivered	
22	05-01-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	UPI	Delivered	

Task 6

Brief Explanation:

- 1) Imported the dataset into Excel.
- 2) Formatted headers and data appropriately.
- 3) Adjusted column widths for readability.
- 4) Applied suitable number/date formats.
- 5) Sorted records in ascending/descending order.
- 6) Applied filters to selected columns.
- 7) Filtered records based on specific conditions.
- 8) Captured screenshots of the final results.

Task 7

Output:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Order_ID ▾	Order_Date ▾	Customer_ID ▾	Customer_Name ▾	Product_ID ▾	Product ▾	Category ▾	Region ▾	Quantity ▾	Unit_Price ▾	Sales ▾	Performance ▾	Payment_Method ▾	Order_Status ▾	
1	ORD1010	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	High	Card	Cancelled	
2	ORD1020	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	High	Net Banking	Cancelled	
3	ORD1030	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	High	Card	Cancelled	
4	ORD1040	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	High	Net Banking	Cancelled	
5	ORD1050	01-06-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	High	Card	Cancelled	
6	ORD1060	01-07-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	High	Net Banking	Cancelled	
7	ORD1070	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	High	Card	Cancelled	
8	ORD1080	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	High	Net Banking	Cancelled	
9	ORD1090	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	₹ 12,500	₹ 62,500	High	Card	Cancelled	
0	ORD1100	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	₹ 12,500	₹ 62,500	High	Net Banking	Cancelled	
1	ORD1002	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	High	Card	Delivered	
2	ORD1012	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	High	Net Banking	Delivered	
3	ORD1022	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	High	Card	Delivered	
4	ORD1032	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	High	Net Banking	Delivered	
5	ORD1042	08-05-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	High	Card	Delivered	
6	ORD1052	07-06-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	High	Net Banking	Delivered	
7	ORD1062	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	High	Card	Delivered	
8	ORD1072	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	High	Net Banking	Delivered	
9	ORD1082	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	₹ 28,000	₹ 56,000	High	Card	Delivered	
0	ORD1092	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	₹ 28,000	₹ 56,000	High	Net Banking	Delivered	
1	ORD1001	05-01-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	High	UPI	Delivered	
2	ORD1011	04-02-2026	C111	Ravi	P001	Laptop	Electronics	West	1	₹ 55,000	₹ 55,000	High	Cash	Delivered	
3	ORD1021	06-03-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	High	UPI	Delivered	
4	ORD1031	05-04-2026	C111	Ravi	P001	Laptop	Electronics	West	1	₹ 55,000	₹ 55,000	High	Cash	Delivered	
5	ORD1041	05-05-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	High	UPI	Delivered	
6	ORD1051	04-06-2026	C111	Ravi	P001	Laptop	Electronics	West	1	₹ 55,000	₹ 55,000	High	Cash	Delivered	
7	ORD1061	05-01-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	High	UPI	Delivered	
8	ORD1071	04-02-2026	C111	Ravi	P001	Laptop	Electronics	West	1	₹ 55,000	₹ 55,000	High	Cash	Delivered	
9	ORD1081	06-03-2026	C101	Arun	P001	Laptop	Electronics	South	1	₹ 55,000	₹ 55,000	High	UPI	Delivered	

Task 7

Output:

ORD1087	24-03-2026	C107	Vijay	P
ORD1097	23-04-2026	C117	Manoj	P
Note:				
Using Countif:				
Delivery Orders Count:			60	
Cancelled Orders Count:			20	
Pending Orders Count:			20	
Using SUMIF():				
total sales for South region:			664000	
Mouse sales in south:			9000	

Performance	
High	
High	
High	
High	
High	
High	
High	
High	
High	
High	

Note:

Determined performance column using SUMIF

Task 7

Brief Explanation:

- Applied **Conditional Formatting** to highlight values based on conditions.
- Used IF() to perform logical checks.
- **Syntax:** =IF(condition, value_if_true, value_if_false)
- Used COUNTIF() to count cells matching a condition.
- **Syntax:** =COUNTIF(range, criteria)
- Used SUMIF() to calculate values based on a condition.
- **Syntax:** =SUMIF(range, criteria, sum_range)
- Verified the results and explained the purpose of each function.

Task 8

OUTPUT:

Order_ID	Order_Date	Customer_ID	Customer_Name	Product_ID	Product	Category	Region	Quantity	Unit_Price	Sales	Performance	Payment_Method	Order_Status
ORD1010	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	High	Card	Cancelled
ORD1020	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	12500	62500	High	Net Banking	Cancelled
ORD1030	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	High	Card	Cancelled
ORD1040	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	12500	62500	High	Net Banking	Cancelled
ORD1050	01-06-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	High	Card	Cancelled
ORD1060	01-07-2026	C120	Deepa	P010	Printer	Electronics	North	5	12500	62500	High	Net Banking	Cancelled
ORD1070	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	High	Card	Cancelled
ORD1080	03-03-2026	C120	Deepa	P010	Printer	Electronics	North	5	12500	62500	High	Net Banking	Cancelled
ORD1090	02-04-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	High	Card	Cancelled
ORD1100	02-05-2026	C120	Deepa	P010	Printer	Electronics	North	5	12500	62500	High	Net Banking	Cancelled
ORD1002	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	High	Card	Delivered
ORD1012	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	High	Net Banking	Delivered
ORD1022	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	High	Card	Delivered
ORD1032	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	High	Net Banking	Delivered
ORD1042	08-05-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	High	Card	Delivered
ORD1052	07-06-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	High	Net Banking	Delivered
ORD1062	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	High	Card	Delivered
ORD1072	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	High	Net Banking	Delivered
ORD1082	09-03-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	High	Card	Delivered
ORD1092	08-04-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	High	Net Banking	Delivered
ORD1001	05-01-2026	C101	Arun	P001	Laptop	Electronics	South	1	55000	55000	High	UPI	Delivered
ORD1011	04-02-2026	C111	Ravi	P001	Laptop	Electronics	West	1	55000	55000	High	Cash	Delivered
ORD1021	06-03-2026	C101	Arun	P001	Laptop	Electronics	South	1	55000	55000	High	UPI	Delivered
ORD1031	05-04-2026	C111	Ravi	P001	Laptop	Electronics	West	1	55000	55000	High	Cash	Delivered
ORD1041	06-05-2026	C101	Arun	P001	Laptop	Electronics	South	1	55000	55000	High	UPI	Delivered

Task 8

OUTPUT:

VLOOKUP:

1) Find the customer name with order id ORD1070

Meena

2) Find the product purchased by the customer with order id ORD1068

Monitor

XLOOKUP:

1) Find the total sales of printer?

62500

2) Customer named meena is from which region?

East

VLOOKUP	XLOOKUP
Searches from left to right	Can search in any direction
Uses a column number	Directly specifies the return column
Requires the lookup column to be first	Lookup and return columns can be separate
Available in older Excel versions	Available in newer Excel versions

Task 8

Brief Explanation:

- Created/used two related Excel tables.
- Identified a common column between the tables.
- Used VLOOKUP() to retrieve related information.
- **Syntax:** =VLOOKUP(lookup_value, table, col_index_num, FALSE)
- Used XLOOKUP() to retrieve related information.
- **Syntax:** =XLOOKUP(lookup_value, lookup_array, return_array)
- Compared the usage of VLOOKUP() and XLOOKUP().
- Captured screenshots of the results.

Task 9

FileHomeWPS PDFInsertPage LayoutFormulasDataReviewViewDesignQueryTell me what you want to do...Share

Clipboard

Font

Alignment

Number

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

Σ

Sort & Find & Filter

Editing

Templates

AI Data Analysis

Export to Image

Split

Merge

Create PDF

Sign PDF

J9

X

✓

fx

14500

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order_ID	Order_Date	Customer_ID	Customer_Name	Product_ID	Product	Category	Region	Quantity	Unit_Price	Sales	Payment_Method	Order_Status	
2	ORD1001	05-01-2026	C101	Arun	P001	Laptop	Electronics	South	1	55000	55000	UPI	Delivered	
3	ORD1002	08-01-2026	C102	Priya	P002	Smartphone	Electronics	East	2	28000	56000	Card	Delivered	
4	ORD1003	11-01-2026	C103	Rahul	P003	Headphones	Electronics	West	3	2500	7500	Cash	Delivered	
5	ORD1004	14-01-2026	C104	Sneha	P004	Desk	Furniture	North	4	12000	48000	Net Banking	Pending	
6	ORD1005	17-01-2026	C105	Karthik	P005	Office Chair	Furniture	South	5	8500	42500	UPI	Cancelled	
7	ORD1006	20-01-2026	C106	Divya	P006	Keyboard	Electronics	East	1	1800	1800	Card	Delivered	
8	ORD1007	23-01-2026	C107	Vijay	P007	Mouse	Electronics	West	2	900	1800	Cash	Delivered	
9	ORD1008	26-01-2026	C108	Ananya	P008	Monitor	Electronics	North	3	14500	43500	Net Banking	Delivered	
10	ORD1009	29-01-2026	C109	Suresh	P009	Bookshelf	Furniture	South	4	6500	26000	UPI	Pending	
11	ORD1010	01-02-2026	C110	Meena	P010	Printer	Electronics	East	5	12500	62500	Card	Cancelled	
12	ORD1011	04-02-2026	C111	Ravi	P001	Laptop	Electronics	West	1	55000	55000	Cash	Delivered	
13	ORD1012	07-02-2026	C112	Pooja	P002	Smartphone	Electronics	North	2	28000	56000	Net Banking	Delivered	
14	ORD1013	10-02-2026	C113	Naveen	P003	Headphones	Electronics	South	3	2500	7500	UPI	Delivered	
15	ORD1014	13-02-2026	C114	Keerthi	P004	Desk	Furniture	East	4	12000	48000	Card	Pending	
16	ORD1015	16-02-2026	C115	Harish	P005	Office Chair	Furniture	West	5	8500	42500	Cash	Cancelled	
17	ORD1016	19-02-2026	C116	Lakshmi	P006	Keyboard	Electronics	North	1	1800	1800	Net Banking	Delivered	
18	ORD1017	22-02-2026	C117	Manoj	P007	Mouse	Electronics	South	2	900	1800	UPI	Delivered	
19	ORD1018	25-02-2026	C118	Swetha	P008	Monitor	Electronics	East	3	14500	43500	Card	Delivered	
20	ORD1019	28-02-2026	C119	Ajaya	P009	Bookshelf	Furniture	West	4	6500	26000	Cash	Pending	

Task 9

PIVOT TABLE

	A	B	C	D	E	F
1	Product-wise Sales by Region					
2						
3	Products					
4		East	North	South	West	Grand Total
5	Bookshelf			130000	130000	260000
6	Desk	240000	240000			480000
7	Headphones			37500	37500	75000
8	Keyboard	9000	9000			18000
9	Laptop			275000	275000	550000
10	Monitor	217500	217500			435000
11	Mouse			9000	9000	18000
12	Office Chair			212500	212500	425000
13	Printer	312500	312500			625000
14	Smartphone	280000	280000			560000
15	Grand Total	1059000	1059000	664000	664000	3446000
16						
17						

Task 9

PIVOT TABLE & PIVOT CHART

